CRISIS INTERVENTION FOR PARAMEDICS

Non-psychotic illness

Marilyn Niffin BSc, ACP
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Learning Outcomes

- Identify the most common causes and/or contributing factors to the development of various mental Health disorders
- Describe the patient assessment findings and behaviours which may be demonstrated in each.
- Describe the therapeutic approaches for each and provide the rationale for each approach

Behavioural Emergencies

- What is a behavioural emergency?
  - *Behaviour is the way a person acts or performs in a given situation*
  - *What is normal? It is important to remember that 'normal' is different from patient to patient.*
  - *What society deems normal is not necessarily absolute*
- When a persons behaviour (usually in response to a stressor) changes, they may put themselves or others around them at risk of harm. This is when it can become a medical emergency
- Very important to remain neutral and unbiased. Non-judgmental approach.

Overview

- Anxiety Disorders
  - *Panic Disorders*
  - *General Anxiety Disorder*
  - *Social Anxiety Disorder*
- OCD
- PTSD
- PNES

Diagnostic and Statistical Manual of Mental Disorders (DSM)

- **DSM–5** is a manual for assessment and diagnosis of mental disorders
- Lists criteria related to specific disorders for use by psychiatrists and health care workers to aid in the diagnosis of mental health disorders

Anxiety Disorders

- GAD and Panic disorder
- OCD (obsessive compulsive disorder)
- PTSD (post-traumatic stress disorder)
- Specific Phobias
- Agoraphobia
- Social Anxiety disorder
- Selective Mutism
- Separation anxiety disorder

Anxiety Related disorders

- Affect 11.6 % of Canadians (aged 18 years or older((Stats Canada, 2013)
- Characterized by dominating apprehension and fear
- Anxiety is
  - *A state of uneasiness, discomfort, apprehension and restlessness*
  - *Often accompanied by physiological symptoms/sensations*

Differential Diagnosis

- Acute Coronary Syndrome
- Alcohol and Substance Abuse Evaluation
- Congestive Heart Failure and Pulmonary Edema
- Costochondritis
- Depression and Suicide
- Hyperthyroidism, Thyroid Storm and Graves Disease
- Hyperventilation Syndrome
- Hypoglycemia
- Mitral Valve Prolapse
- Myocardial Infarction
- Pneumonia, Pneumothorax, Pneumomediastinum
- Pulmonary Embolism
- Schizophrenia
- Toxicity, drugs, Withdrawal Syndromes

Anxiety Related Disorders

- Etiology: Hypothesis
  - *Disturbances in the limbic system and other parts of the brain and the connections between these structures*
  - *Excess catecholamine release*
- 30-50% of persons with anxiety disorder also suffer from agoraphobia
  - *Fear of entering crowded or open public spaces (experiencing a panic attack in a public place)*
- 50 – 60% also suffer from major depressive disorder

Anxiety disorders

- **Panic disorders**
  - *Recurrent and unexpected panic attacks followed by at least* ***one month of****: constant concern about having another panic attack or the consequences of having an attack (I.e. heart attack) and significant behaviour changes related to the attacks (avoiding activities in fear of an attack)*
- Panic attacks are sudden feelings of extreme anxiety and great emotional distress
- Unprovoked and peak within 10 min
- Usually lasts more than 1 hour
- Panic Attacks do not necessarily mean you have a panic disorder
- Peak in late adolescence and in middle age (forties)
- Can occasionally be difficult to differentiate from cardiac or respiratory conditions

Panic Disorders

- Etiology: genetically inherited neurochemical dysfunction
- Common co-morbidities
  - *COPD, IBS, migraines, OCD, restless leg syndrome, specific phobias, social anxiety disorder, agoraphobia*
  - *Cardiovascular disorders (mitral-valve prolapse, HTN, cardiomyopathy, stroke)*
- ***Panic attack patients are twice as likely to develop coronary artery disease***

Panic Disorders - Signs and Symptoms

- **Physical Symptoms**
  - Dyspnea, palpitations, tachycardia, chest pain, dizziness, numbness, trembling, sweating, hyperventilation, chills, nausea, abdominal pain, feelings of choking
- **Psychological/cognitive symptoms**
  - Expressed fear of dying, impending doom, losing control, derealization, depersonalization
- **Behavioural Symptoms**
  - Hyperkinesis, pressured speech, exaggerated startle response

Panic Attack - Triggers

- Injury (eg, accidents, surgery), Illness
- Interpersonal conflict or loss
- cannabis (? breath-holding)
- Stimulants e.g caffeine, decongestants, cocaine
- Certain settings e.g. , such as stores and public transportation (agoraphobia)
- Sertraline, which can induce panic in previously asymptomatic patients
- The selective serotonin reuptake inhibitor (SSRI) discontinuation syndrome

Panic Disorders - Assessment

- Scene – Ensure personal safety
- Primary Assessment
  - *Differential diagnosis – suspect life-threatening emergencies*
  - *Support ABC's*
  - *General Impression*
- Secondary Assessment
  - *Rule out medical related causes*

Panic Disorders - Management

- Listen
- Spend Time
- Reassure
- Do not threaten
- Place yourself at the patients level (keep safety in mind) you do not want to appear intimidating
- Keep eye contact as appropriate
- Avoid being judgmental
- NEVER lie to the patient

Panic Disorders - Management

- DO NOT begin to coach any increased respiratory rate (hyperventilation) until medical issues have been ruled out
- O2 PRN
- No CO2 re-breathing (paper bag)
- IF safe, coach breathing to reduce rate and volume
- If in doubt, treat as a medical condition and transport
- Treatment Includes:
  - *CBT, medication, relaxation training, meditation, MBSR, stress management, family therapy*

Generalized Anxiety Disorder

- Characterized by excessive anxiety and worry occurring on more days than not for at least 6 months
- Affects females twice as much as males
- Approximately 50% of cases begin in childhood or adolescence
- Typically runs a fluctuating course – symptoms are more prominent when associated with other life stress
- Most patients with anxiety disorder have another Mental Health problem (phobias, social anxiety, major depression, mood disorders)
- Often associated with alcohol, substance abuse and personality disorders

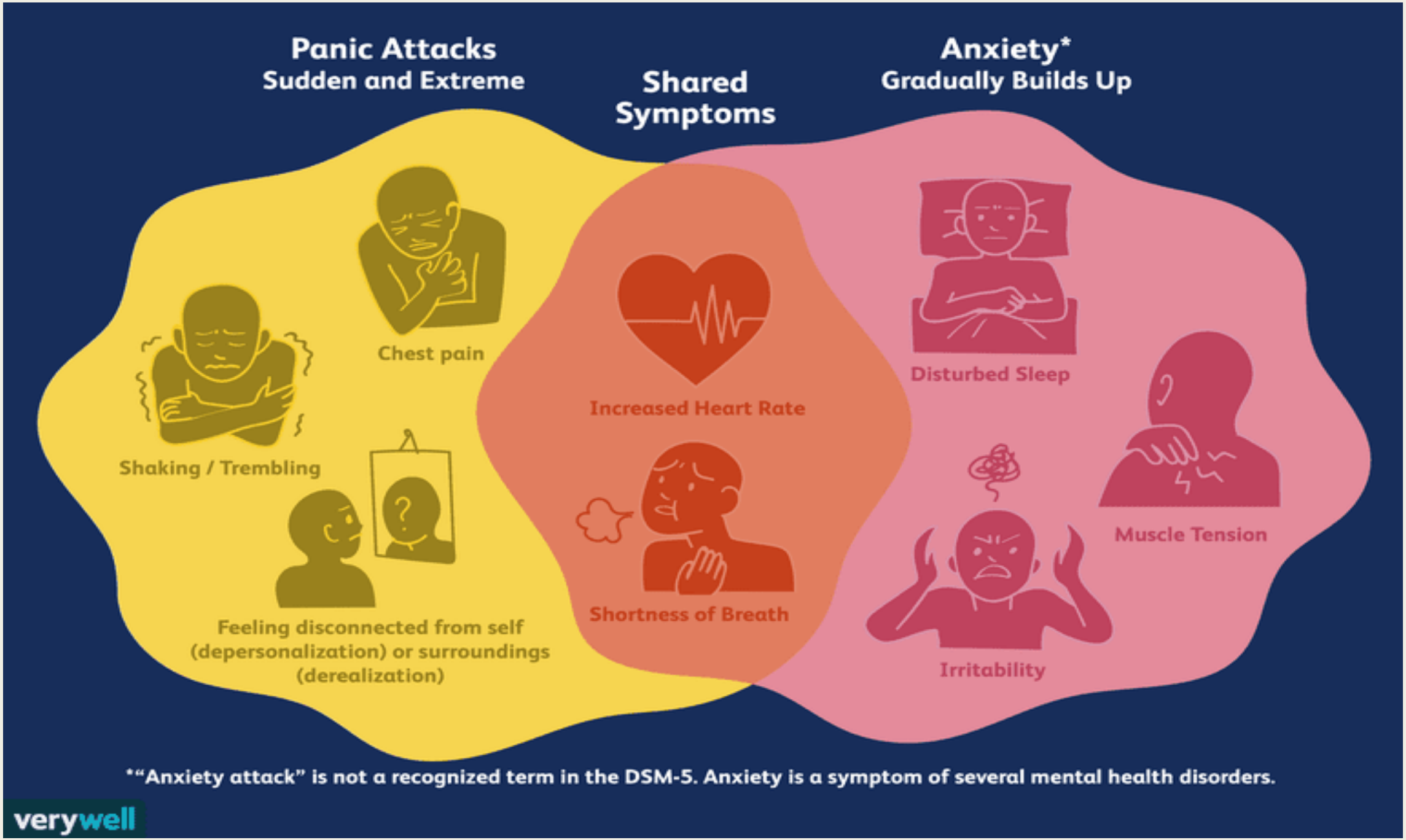
GAD – risk factors

- Similar to other anxiety disorders
  - *Abuse, family history (parental mental health disorder, low income*
- In combination with other biological and psychological disorders
- Challenging or life altering situations

General Anxiety Disorder

Diagnostic Criteria for GAD – DSM IV

- a) Excessive anxiety and worry (apprehensive expectation), occurring more days than not for at least 6 months
  - *about a number of events or activities (such as work or school performance).*
- b) The person finds it difficult to control the worry.
- The anxiety and worry are associated with three (or more) of the following six symptoms (with at least some symptoms present for more days than not for the past 6 months). Note: Only one item is required in children.
  - *restlessness or feeling keyed up or on edge*
  - *easily fatigued*
  - *difficulty concentrating or mind going blank*
  - *irritability*
  - *muscle tension*
  - *sleep disturbance*


Panic Attacks
Sudden and Extreme
Shared Symptoms
Anxiety*
Gradually Builds Up
Chest pain
Shaking / Trembling
Feeling disconnected from self (depersonalization) or surroundings (derealization)
Increased Heart Rate
Shortness of Breath
Disturbed Sleep
Muscle Tension
Irritability
*"Anxiety attack" is not a recognized term in the DSM-5. Anxiety is a symptom of several mental health disorders.
verywell

General Anxiety Disorder

- Medications
  - *Antidepressants; benzodiazepines, beta-blockers, Buspirone*
  - *Drug choice is symptom dependent and the patients ability to tolerate side effects (different with each classification of drug*

GAD - Medications

Antidepressants

- Paroxetine (paxil)
- Venlafaxin HCL (Effxor XR)
  - *Most commonly prescribed for GAD however other antidepressants may be effective and may be used*

Benzodiazepines

- Alprazolam (Xanax)
- Chlordiazepoxide (Librium®)
- Clonazepam (Klonopin®)
- Diazepam (Valium®)
- Lorazepam (Ativan®)

GAD - Medications

Beta Blockers

- used to treat cardiac conditions
- may also be used in anxiety disorders to stop the physical symptoms of anxiety
  - *reduce nervous tension, sweating, panic, high blood pressure, and shakiness.*
- not typically considered a first-line treatment

Buspirone

- attributes include the following:
  - *Does not interact with alcohol (but not recommended to consume together)*
  - *Few or no withdrawal symptoms*
  - *Lack of relaxant muscular effect*
  - *Lack of sedative effect*
  - *Works within 3 to 6 weeks and works effectively*
- side effects
  - *Agitation, excitement confusion dizziness drowsiness, fatigue, weakness, headaches nausea nervousness*

GAD - Medications

- The side effects associated with discontinuation of benzodiazepine (and others) can make stopping medications difficult
- can experience withdrawal syndrome

Social Anxiety Disorder

- AKA social phobia
  - *Fear of anxiety about being humiliated or scrutinized in social situations (lasts at last 6 months)*
  - *Fear causes significant distress or impairment in day-to-day functioning*
    - Associated with social interactions – i.e. meeting new people, dating, participating in small groups, playing sports
  - *"People will think I'm weird"*
  - *Selective Mutism often associated with Social anxiety disorder*
- Signs and Symptoms include:
  - *Blushing, sweating, dry mouth, rapid heart beat, trembling or shaking, chest tightness*

Social Anxiety Disorder

- Most effective treatment is CBT
- Anti-anxiety and antidepressants also effective
- Support groups
- Exercise
- Healthy eating
- Managing stress
- meditation

Obsessive – Compulsive Disorder (OCD)

- Persistent involuntary thoughts that provoke anxiety and anxiety management rituals
- Patients typically avoid disclosing symptoms

Clinical Manifestations

- **Obsessions** are strong, persistent, intrusive, uncontrollable thoughts
- **Compulsions** are repetitive, ritualistic actions performed with urgency, typically are related to obsessions
- Persons affected are unable to stop, govern, or resist obsessions and compulsions
  - *4 consistent syndromes:* ***Symmetry/ordering; contamination/cleaning; hoarding; obsessive checking***
- Uncertainty is the hallmark of compulsions
  - *Results from a discrepancy between sensory information and internal beliefs*

OCD

- Strong genetic component – inherited risk
- Treatment includes:
  - *Exposure therapy*
  - *CBT in conjunction with medications*
  - *Antianxiety, antidepressants, antipsychiatrics, and beta-blockers, tricyclic antidepressants (TCA) and anticonvulsants also helpful*

PTSD –Post Traumatic Stress Disorder

- Precipitated by a traumatic event
- Characterized by anxiety, depression, and dissociation are common
- **Acute Stress disorder** occurs up to 1 month after the event – after 1 month without resolution of symptoms it becomes PTSD
- **Acute PTSD** - symptoms less than three month
- **Chronic PTSD** - symptoms more than three months
  - *Although symptoms usually begin within 3 months of the event, delayed onset (months, even years) after the event is possible*
- Strong correlation between PTSD and suicidal ideation

PTSD- defined

- The trigger to PTSD is exposure to actual or threatened death, serious injury or sexual violation. The exposure must result from one or more of the following scenarios, in which the individual:
  - *directly experiences the traumatic event;*
  - *witnesses the traumatic event in person;*
  - *learns that the traumatic event occurred to a close family member or close friend (with the actual or threatened death being either violent or accidental); or*
  - *experiences first-hand repeated or extreme exposure to aversive details of the traumatic event (not through media, pictures, television or movies unless work-related).*

PTSD

- The disturbance, regardless of its trigger, causes clinically significant distress or impairment in the individual's social interactions, capacity to work or other important areas of functioning. It is not the physiological result of another medical condition, medication, drugs or alcohol.

PTSD risk factors

- PTSD can affect anyone
- An estimated 1 of 10 women will develop PTSD
- Victims of physical and sexual assault face the greatest risk of developing PTSD
- women are about twice as likely to develop PTSD
- more likely to experience trauma that involves interpersonal violence, including rape and severe beatings.
- Victims of domestic violence and childhood abuse also are at tremendous risk for PTSD

PTSD

Signs and Symptoms

- Intrusive symptoms include flashbacks, dreams, recurring thoughts of the event
- Negative cognitive and mood changes
- Avoids all stimuli associated with trauma
- Self-destructiveness, hypervigilance, exaggerated startle response
- ***Often reluctant to accept help or acknowledge a problem***
- Stigma

PTSD - Treatments

- Medications – SSRI; Adrenergic blockers
- CBT
- Reframing
- Gradual exposure to the trauma with imagination, real-life or VR therapy
- Cognitive restructuring
- Relaxation techniques
- Eye movement desensitization and reprocessing

PNES

- Psychogenic Non-epileptic Seizures

Seizure (noun)

- 1. a: the act, action, or process of seizing : the state of being seized

  b: the taking possession of person or property by legal process

- 2. a: a sudden attack (as a disease)

  especially : the physical manifestations (such as convulsions, sensory disturbances, or loss of consciousness) resulting from abnormal electrical discharges in the brain (as in epilepsy)

  b: an abnormal electrical discharge in the brain

Merriam-Webster Dictionary

Psychogenic → "Having a psychological origin or cause, rather than a physical one."

Non-Epileptic → "...not caused by abnormal activity in the brain."

Seizure → "The sudden attack or recurrence of a disease"

What is PNES

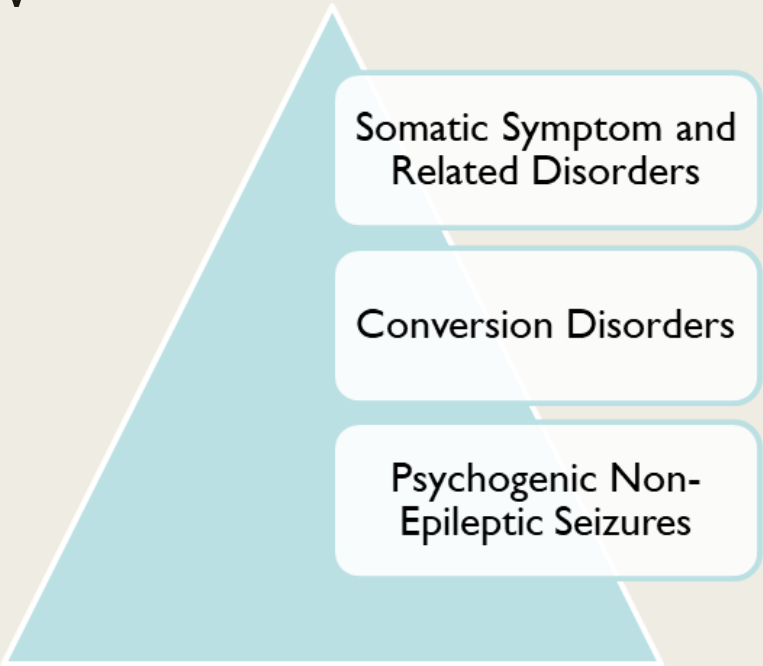
- Psychogenic Non-Epileptic Seizures are seizure activity that does not originate from electrical disturbance, but from a psychological condition
-
- Classified as conversion disorder in the DSM-5
- May also be referred to as:
  - *Conversion Disorder*
  - *Non Epileptic Attack Disorder (NEAD)*
  - *Pseudo Seizure****
  - *Pseudo Non-Epileptic Seizure (Also PNES)*
  - *Dissociative Seizure*

Conversion Disorder – Diagnostic Criteria

- A. One or more symptoms of altered voluntary motor or sensory function.
- B. clinical findings provide evidence of incompatibility between the symptom and recognized neurological or medical conditions
- C. The symptom or deficit is not better explained by another medical or mental disorder.
- D. The symptom or deficit causes clinically significant distress or impairment in social, occupational, or other important areas of functioning or warrants medical evaluation

Diagnosing PNES

- Video-EEG is the best method of diagnosing PNES
- It must first be determined if a seizure is related to neuronal activity
- If not related to medical or neurological issues, PNES may be suspected and diagnosed with DSM-V



Risk Factors

- History of childhood abuse and neglect
- Stressful life events
- Neurological disease (**PNES is common in patients who also have epilepsy**)
- Anxiety (especially panic disorder)
- Depression
- Personality disorders
- Sex: occurs more often in females than males
- Family history of epilepsy (Motor movement in PNES activity may be ‘learned’)

PNES Presentation

- PNES episodes present as epileptic seizures
- Adult Non-Epileptic Seizure
- Adult Non-Epileptic Seizure #2
- Pediatric Non-Epileptic Seizure
- May have preceding auras or warning signs
- Motor symptoms: weakness or paralysis; abnormal limb posturing
  - *These may appear as asymmetrical in PNES; e.g. shaking head side to side*
- Sensory symptoms: altered, reduced, or absent skin sensation, vision, or hearing
- Episodes of abnormal generalized limb shaking with apparent impaired or loss of consciousness resembling epileptic seizures
  - **In attacks resembling epilepsy or syncope, the occurrence of closed eyes with resistance to opening*

Distinguishing PNES from Epilepsy

- What may be observed while responding:
  - *Apparent resistance to eye opening (including passive)*
  - *Asymmetrical motor activity (side to side head shaking)*
  - *Often no apparent post-ictal phase (or inconsistent history of)*
  - *Symptoms may be manipulated or influenced by bystander*
- What may be observed in hospital
  - *Normal ictal and postictal video-EEG*
  - *No, or low changes in prolactin serum levels (more significant increases in Epilepsy)*
- It is important to remember that individuals with PNES may also have epilepsy. It may also be difficult to distinguish a non-epileptic seizure from an epileptic seizure.

Managing PNES

Seizure Standard

In situations involving a patient in seizure (or post-ictal), the paramedic shall:

1. consider potential life/limb/function threats and/or underlying disorders, such as,
   a. intracranial event,
   b. hypoglycemia,
   c. in pregnant patients or recent post-partum patients, eclampsia,
   d. in patients ≥50 years of age with new onset or recurrent seizures,
      i. brain tumour or other intracranial event (*e.g.* hemorrhage, thrombosis),
      ii. cardiac dysrhythmias,
      iii. cardiovascular disease,
      iv. cerebrovascular disease, and
      v. severe hypertension,
   e. in neonates,
      i. traumatic delivery,
      ii. congenital disorders,
      iii. prematurity, and
      iv. hypoglycemia,
   f. in young children febrile convulsions associated with infection,
   g. infection (*e.g.* central nervous system, meningitis),
   h. alcohol withdrawal (including delirium tremens)
   i. drug ingestion/withdrawal, and
   j. known seizure disorder;
2. if patient is in active seizure,
   a. attempt to position the patient in the recovery position,
   b. attempt to protect the patient from injury, and
   c. observe for,
      i. eye deviation,
      ii. incontinence,
      iii. parts of body affected, and
      iv. type of seizure (*e.g.* full body, focal);
3. perform, at a minimum, a secondary survey to assess,
   a. for seizure-related occurrences, such as,
      i. bleeding from the mouth,
      ii. incontinence,
      iii. secondary injuries resulting from the seizure, and
      iv. tongue injury; and
4. prepare for potential problems, including,
   a. airway compromise,
   b. recurrent seizures, and
   c. post-ictal combativeness or agitation.

Managing PNES

- Important information to gather from these patients:
  - *Triggers for their seizures?*
  - *Was there warning or aura before this seizure?*
  - *Description of seizure activity?*
  - *Duration and intensity of activity?*
  - *Changes throughout episode?*
  - *How quickly did the patient return to normal?*
  - *Do these episodes occur in multiples?*
  - *Past hospitalization following seizures?*
  - *History of seizure activity?*
  - *Diagnosis?*
  - *Treatment?*

Managing PNES

- LISTEN to your patient
- What they are experiencing is REAL
- Focal /partial vs Generalized seizures
- If you are unable to distinguish the seizure-like activity between PNES and epileptic, treat as per the seizure standard (BLS PCS)
- Many patients will be treated with medication and as epileptic until properly diagnosed with video EEG

Questions???